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Review

Integrating Artificial Intelligence into Smart Infrastructure Management for Sustainable Urban Planning

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Abstract

This paper systematically reviewed studies on the integration of Artificial Intelligence (AI) into infrastructure management to support sustainable urban planning across three primary domains: predictive maintenance and energy optimization, traffic and mobility systems, and public participation with ethical considerations. Findings from thirty peer-reviewed studies underscore how AI-driven models enhance operational efficiency, sustainability, and governance in smart cities. Effective management of AI-driven smart infrastructure can transform urban planning by optimizing resources efficiency and predictive maintenance, including 15% energy savings, 25–30% cost reductions, 25% congestion reduction, and 18% decrease in travel times. Similarly, participatory digital twins and citizen-centric approaches are found to enhance public participation and help address ethical issues. The findings further reveal that AI-based predictive maintenance frameworks improve system reliability, while deep learning and hybrid models achieve up to 92% accuracy in traffic forecasting. Nonetheless, obstacles to equitable implementation, including the digital divide, privacy infringements, and algorithmic bias, persist. Establishing ethical and participatory frameworks, anchored in responsible AI governance, is therefore vital to promote transparency, accountability, and inclusivity. This study demonstrates that AI-enabled smart infrastructure management strengthens urban planning by enhancing efficiency, sustainability, and social responsiveness. It concludes that achieving sustainable and socially accepted smart cities depends on striking a balance between technological innovation, ethical responsibility, and inclusive governance.

Keywords: artificial intelligence; smart cities; town planning; smart infrastructure; urban sustainability; SDG 11

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1. Introduction

Urbanization is increasing at an alarming level worldwide, exerting enormous pressure on scarce resources, infrastructure, and urban governance [1,2]. Over half of the world's population currently resides in urban areas, which is anticipated to nearly reach 70% by 2050 [3]. This pace of urban growth is more pronounced in the Global South, where urban expansion overwhelms infrastructure development [4,5]. This rapid growth poses significant urban challenges, including pressure on existing services and sustainability concerns [6].

The impacts of urbanization can be felt through environmental and socio-economic crises. While urban areas face severe levels of pollution, traffic congestion, and resource depletion [7–9], socioeconomic inequalities continue to widen, leaving many urban residents without adequate access to clean water, sanitation, shelter, and healthcare [10].

The environmental toll is equally noteworthy, as urban areas contribute substantially to global greenhouse gas (GHG) emissions due to intensive energy consumption [11,12]. Also, the growing frequency of natural disasters might exacerbate these challenges, straining already fragile urban management [13].

The complexity and scale of these challenges have rendered traditional urban planning models insufficient to deliver public service, manage infrastructure, allocate resources, and protect the environment [14]. These limitations have prompted urban planners to explore innovative solutions, with Artificial Intelligence (AI) evolving as a transformative tool for addressing such multifaceted urban challenges [15].

As such, urban areas worldwide are increasingly adopting AI to revolutionize governance by enabling more adaptive, efficient, and resilient urban systems capable of addressing challenges such as resource shortages, traffic congestion, and energy inefficiencies [16–18]. As it receives and processes immense quantity of real-time data, AI can facilitate infrastructure management, predictive maintenance, optimize transport, and enhance energy management beyond what traditional methods can achieve [19,20]. Consequently, AI-driven infrastructure management is essential in achieving efficiency and urban sustainability [21].

Among the key advantages of AI is predictive maintenance, where algorithms analyze sensor data to anticipate system failures before they happen, thus reducing operational costs, preventing disruptions in vital services such as water, electricity, and transportation networks [20,22]. Similarly, AI-based traffic management systems can alleviate congestion by analyzing real-time traffic data and dynamically adjusting signal timings to minimize travel times and GHG emissions [23].

Also, AI plays a crucial role in urban energy optimization and sustainable energy ecosystems, where smart grids equipped with AI can balance energy supply and demand in real time, minimizing waste and integrating renewable energy sources to lower emissions, manage energy storage, and provide predictive insights into future energy needs [24]. Moreover, AI applications for citizen engagement have shown promise in improving inclusivity and accessibility, although implementation challenges persist [25–27].

Despite its growing interest, significant gaps remain in understanding the full transformative potential of AI in urban planning. While numerous studies explore its applications in specific areas such as traffic management and energy optimization, few comprehensively assess the broader impact of AI across interconnected infrastructure systems [28]. Questions also remain regarding how AI can be well integrated into existing urban frameworks and the barriers impeding its widespread adoption [20,29]. Ethical challenges related to data privacy, security, and transparency further complicate implementation [30]. The risk that AI could deepen existing inequalities in access to smart infrastructure, particularly in developing regions, also warrants critical investigation [31].

Similarly, resilience as the ability of urban systems to adapt, recover, and maintain functionality under stress has merged as a crucial outcome of effective AI integration [32]. Although resilience is intrinsically linked to the optimization and predictive capabilities of AI across various infrastructure systems, the literature discussed them separately [33]. This study, therefore, views resilience not as an isolated goal but as a cumulative outcome of AI-based improvements in maintenance, resource management, and mobility systems. However, ensuring that AI systems remain robust and adaptive during crises is a persistent challenge.

In participatory urban planning, AI systems can promote inclusivity and transparency, supporting data-driven and equitable decision-making [34,35]. Yet, digital divide, data privacy, ethical considerations and equitable access are major obstacles [36,37]. Address-

ing these concerns is crucial for ensuring that AI-driven urban innovations foster more sustainable and inclusive development.

These knowledge gaps call for a systematic review to assess the efficacy of AI-based smart infrastructure across diverse urban contexts and to identify best practices for integrating AI into planning frameworks. Accordingly, this study aims to investigate AI-driven smart infrastructure management within urban planning, focusing on three core themes: (i) predictive maintenance and resources and energy optimization; (ii) traffic and mobility management; and (iii) public participation and ethical considerations.

While proponents underscore the ability of AI technologies to optimize infrastructure, minimize inefficiencies, and enhance sustainability [16], critics caution that technological solutions alone cannot address socio-economic inequalities [30]. Moreover, ethical concerns about data privacy, algorithmic bias, and accountability demand careful attention. The present study contributes in three major ways. First, it provides a structured synthesis of technical and social dimensions of AI in urban planning. Second, it highlights ethical and equity considerations that are often overlooked in existing literature. Third, it offers practical insights to guide policymakers and urban planners in adopting AI responsibly.

To guide this review, the following research question is posed: How can AI-driven smart infrastructure management transform urban planning across the three key themes? This question anchors the exploration of the role of IA in shaping more adaptive, efficient, and inclusive urban future.

The rest of the paper is structured as follows: Section 2 outlines the conceptual framework underpinning this study, underscoring the three key themes of AI-driven smart infrastructure management. Section 3 explains the methodology and systematic review process following the PRISMA approach. Section 4 details the study's results, followed by Section 5, which analyzes the main findings and their implications. Lastly, Section 6 wraps up the paper with core recommendations.

2. Literature Review

This section synthesizes findings from the literature review and the conceptual framework to establish a structured foundation for understanding how AI can contribute to smart infrastructure management in urban areas, which are increasingly complex and evolving due to rapid population growth. As such, AI-driven technologies are pivotal in enhancing the efficiency, sustainability, and resilience of urban infrastructure, particularly in domains such as predictive maintenance and resources optimization, traffic and mobility management, and public participation and ethical considerations [20,25]. Through the processing of massive real-time data, AI can facilitate proactive decision-making and operational efficiency, surpassing the traditional infrastructure management approach. However, the transformative potential of AI technologies extends beyond technical innovation by introducing new layers of ethical ang governance complexity, particularly in areas concerning data privacy and equitable access [30].

Predictive maintenance exemplifies AI's ability to monitor urban infrastructure in real time, anticipate potential failures, optimize resource allocation and extend the operational lifespan of critical urban assets such as roads, bridges, utilities and energy grids [38]. Beyond proactive maintenance, studies demonstrate substantial improvements in urban efficiency. Machine learning, IoT, and smart grid technologies have achieved remarkable results, including 15% energy savings equivalent to 1.6 TWh annually, while investments of $1.2 billion have prevented $3.7 billion in flood damages [37]. Advanced forecasting models using neural networks and gradient boosting have reached coefficient of determination values of 0.9835, optimizing 1.09 million kWh of electricity consumption [39]. Infrastructure maintenance has been revolutionized through AI applications, with studies reporting 30%

faster response times, 25% productivity gains, and 92% accuracy in predictive maintenance systems [40,41].

Traffic management, on the other hand, involves AI-driven systems that optimize urban mobility through real-time data analysis, reducing congestion, travel times, and emissions [23]. Traffic and mobility management applications showcase equally impressive outcomes. Deep learning and computer vision techniques have reduced traffic congestion by up to 25% and decreased travel times by 18% [42]. Predictive accuracy for traffic management and accident detection consistently exceeds 90%, with some studies achieving 98% accuracy in vehicle detection systems [43,44].

Public participation refers to citizens' involvement in urban decision-making processes, whereas ethical consideration denotes the attention to ethical issues such as data privacy, algorithmic bias, and equitable access. However, these technological advances raise critical ethical and participatory concerns. Studies emphasize the importance of addressing digital divides, algorithmic bias, and privacy protection through participatory digital twins and inclusive design approaches [26,45]. This comprehensive body of research demonstrates that AI-driven smart infrastructure can enhance urban planning by creating more efficient, sustainable, and socially responsive urban environments. In the context of AI-driven infrastructure management, these aspects emphasize not only how digital and AI-enabled platforms can support more inclusive participation, but also the broader transformative potential and risks of AI in urban settings [38,46].

Conceptual Framework of AI-Driven Smart Infrastructure Management

A conceptual framework (Figure 1) was developed to visualize the interconnected domains, challenges, and sustainability goals of AI-driven smart infrastructure management. The framework centers on three key domains through which AI technologies are transforming urban systems: (1) predictive maintenance and resource optimization, (2) traffic and mobility management, and (3) public participation and ethical considerations. Each domain is linked to cross-cutting challenges, including data privacy, algorithmic bias, and equitable access, while contributing to sustainability outcomes like resource efficiency, decarbonization, and resilience [20,47].

In the framework, the solid arrows represent direct, unidirectional relationships between specific domains and their corresponding sustainability goals, whereas the dotted lines denote interdependencies without fixed directional flow. These interconnections emphasize that these three domains are interconnected, with advancements in one domain influencing and supporting others. For example, predictive maintenance fosters infrastructure reliability, which in turn enhances traffic flow and contributes to optimal resource use. Also, optimized resource systems can help reduce operational strains on infrastructure, bolstering predictive maintenance efforts.

Likewise, AI-driven traffic and mobility management systems, including adaptive traffic signals and real-time congestion prediction tools can enhance urban mobility while simultaneously reducing carbon emissions, thereby strengthening both resource efficiency and infrastructure reliability [48,49]. In parallel, AI-supported public participation platforms, such as digital feedback systems and participatory decision-making dashboards, foster transparency and public trust, ensuring that infrastructure solutions are responsive to community needs. Addressing ethical dimensions, such as addressing algorithmic bias and equitable access to AI-enabled services, is vital in ensuring that the integration of AI into infrastructure management advances fairness and inclusivity [38,50]. Furthermore, the solid arrows within the framework illustrate the direct relationship among the three domains and broader sustainability outcomes, including resource efficiency, reduced carbon emissions,

and infrastructure resilience, demonstrating how progress in each domain collectively contributes to achieving these broader goals.

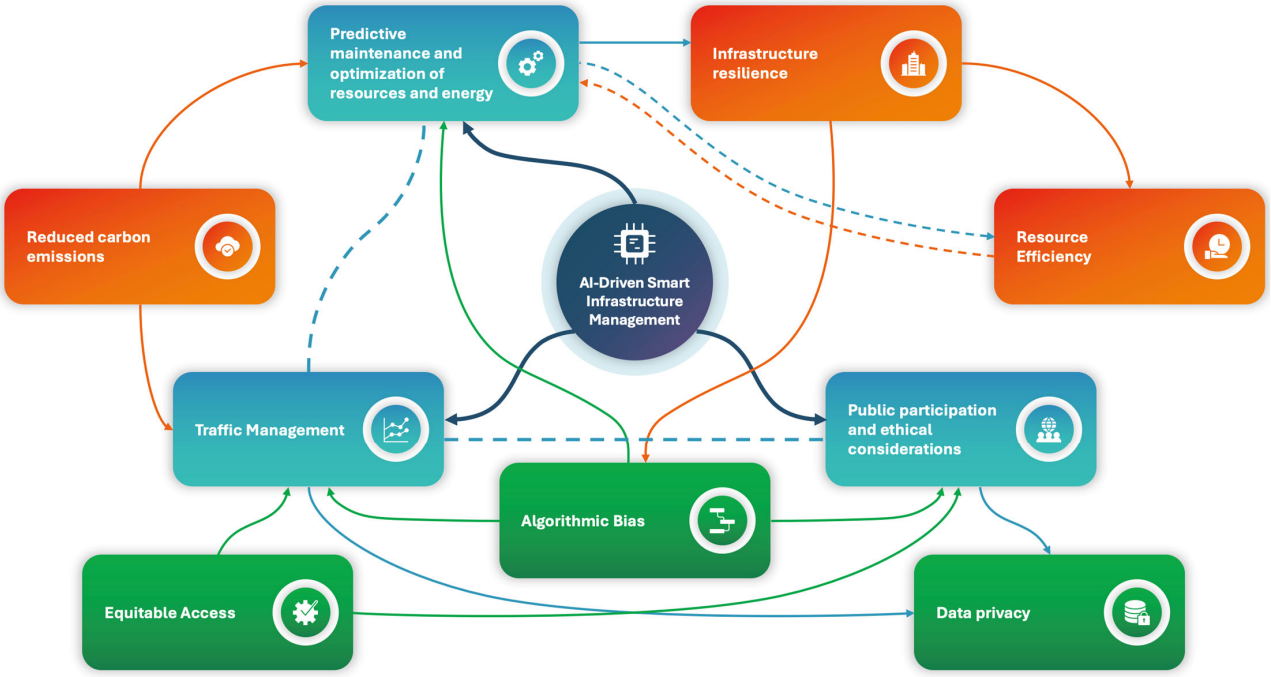


Figure 1. Conceptual Framework of AI-Driven Smart Infrastructure Management.

Table 1 below offers a detailed breakdown of each domain, including key AI applications, their specific impacts on urban systems, and the challenges encountered. The table outlines how AI is applied in urban infrastructure, providing a more detailed view of how each domain contributes to the overall transformation of urban environments.

Table 1. Scope of AI-Driven Smart Infrastructure Management.

Domain	Definition	AI Applications	Impact on Urban Systems	References
Predictive maintenance and optimization of resources and energy	Use of AI to predict infrastructure failures through sensor data and analytics, alongside enhanced management of urban resources and energy systems, particularly smart grids and related infrastructure.	AI systems analyze sensor data from infrastructure components (e.g., roads, bridges, utilities) and balance the supply and demand of urban resources and energy, particularly by optimizing energy use and integrating renewable sources.	This domain reduces operational costs, prevents service disruptions, and extends the lifespan of infrastructure. It also increases overall efficiency, minimizes waste, and lowers emissions through optimized resource and energy use, typically measured in terms of energy savings (MWh), reduction in network losses, and CO_2 emission reductions.	Hector & Panjanathan [38]; Verma et al. [24]; Dellosa & Palconit [47]; Wolniak & Stecuła [20]; Cina et al. [51]
Traffic and mobility management	Use of AI to optimize traffic flow and urban mobility in real time.	AI-driven traffic management systems enable dynamic traffic signal control and congestion prediction using real-time data analytics.	These systems reduce congestion, lower travel times, and decrease vehicle emissions, thereby improving overall urban mobility. Performance is commonly measured through average travel time reduction, congestion index improvements, and reductions in vehicle-related CO_2 emissions.	Barth & Boriboonsomsin [23]; Ullah et al. [19]; Ismaeel et al. [52]

Table 1. *Cont.*

Domain	Definition	AI Applications	Impact on Urban Systems	References
Public participation and ethical considerations	Public participation and ethical considerations encompass citizen engagement in decision-making alongside concerns around data privacy, fairness, and equitable access to AI technologies in cities.	AI-enabled platforms for citizen feedback (e.g., digital reporting apps, participatory dashboards), online deliberation tools that facilitate inclusive decision-making, and regulatory frameworks ensuring algorithmic transparency, fairness, and accountability.	These approaches strengthen transparency, trust, and inclusiveness in urban governance while addressing risks related to data privacy and algorithmic fairness. Their impact can be evaluated using participation rates in digital platforms, citizen satisfaction surveys, and fairness/equity indices in access to services.	Goodman & Flaxman [30]; Kitchin [46]; Sánchez & Brenman [25]; Berigüete et al. [26]

3. Materials and Methods

The present study conducted a systematic review of literature on AI-driven smart infrastructure management. The protocol was designed based on the procedures recommended by Kitchenham and Charters [53] and following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines, ensuring a rigorous and transparent review process [54]. The Scopus database was selected for this review due to its wide coverage of research related to AI, urban planning, and infrastructure management. This database provides a broad scope of peer-reviewed articles, ensuring the inclusion of interdisciplinary studies spanning fields such as computer science, engineering, and urban studies. Additionally, gray literature, including government reports and industry white papers, was considered to complement academic sources with insights from real-world applications of AI technologies.

The search strategy captured both the technical and applied dimensions of AI in urban infrastructure. To refine the scope and ensure a thorough but focused retrieval of studies, a carefully constructed set of search terms was used. These included combinations of key terms such as “artificial intelligence” and “AI”, paired with terms like “smart infrastructure”, “urban infrastructure”, “predictive maintenance”, “traffic management”, “energy optimization.”, “public participation” and “ethical considerations”. These search terms were applied using Boolean operators (AND, OR) to refine the search results and guarantee the inclusion of studies specifically focused on the intersection of AI and urban infrastructure. The complete search strategies are presented in Appendix A. The inclusion of articles up to 2025 ensures the review incorporates the most current research and developments in AI-driven infrastructure. To ensure a global perspective, no geographical limitations were applied. The inclusion criteria were designed to focus on studies that directly addressed the key areas of AI-driven smart infrastructure management as presented in Table 2.

The literature selection process followed a multi-stage approach to ensure that the review included only papers that are high-quality, and most related to the topic. Initially, the downloaded files were exported to Zotero (Version 6.0.27), where duplicates were merged and the titles and abstracts of the studies were screened to remove irrelevant or out-of-scope articles. This initial stage helped filter out papers that did not meet the basic criteria for inclusion, such as those focused on unrelated technologies or non-urban applications of AI.

After the initial screening, the full texts of the remaining studies were comprehensively reviewed to evaluate their significance to the research objectives. At this stage, studies were evaluated according to the depth of their analysis, their contribution to the understanding of the role of AI in urban infrastructure management, and whether they provided empirical or case-based evidence. Studies that passed this stage were further categorized into themes based on their primary focus area predictive maintenance and optimization

of resources and energy, traffic and mobility management, and public participation with ethical considerations.

Table 2. Inclusion and Exclusion Criteria.

Category	Inclusion Criteria	Exclusion Criteria
Scope	Studies focused on urban planning, smart cities, or urban infrastructure management; conducted in urban settings	Studies focusing solely on rural or non-urban areas; research limited to individual buildings; AI unrelated to urban planning.
Technology focus	Research involving AI applications (ML, DL, IoT, Big Data, Digital Twins, Robotics).	Studies using only traditional or non-AI planning methods; basic digitization; hardware-only systems.
Themes	Must address at least one theme: (1) Predictive maintenance and resource/energy optimization; (2) Traffic and mobility management; (3) Public participation and ethical considerations.	Studies not addressing any of the defined themes (predictive maintenance, mobility, or ethics/public engagement).
Study design	Empirical (quantitative, qualitative, or mixed-methods), systematic review, case studies and theoretical.	Opinion pieces; unclear or unvalidated methodology; insufficient AI details or outcomes.
Publication criteria	Published 2018–2025; peer-reviewed or Peer-reviewed journals or recognized conference proceedings; book chapters.	Studies published before 2018.
Language and accessibility	English-language, full-text available.	Studies without measurable outcomes or accessible full text.

To safeguard transparency and replicability, the entire process was conducted by two reviewers, who independently assessed the studies and cross-checked the results. Disagreements between reviewers were addressed through dialogue, and when needed, a third reviewer was consulted to help achieve consensus. The selection process is subject to certain limitations. First, the review included only English-language publications, excluding papers in other languages. Second, the search was limited to Scopus, the largest academic database.

The initial search across databases yielded a total of 473 studies. Titles and abstracts of the studies were evaluated for relevance according to the predetermined inclusion and exclusion criteria Following the abstract screening, a total of 177 studies were selected for full-text review. During this phase, each study was carefully examined to determine its relevance to the research objectives, with a particular focus on empirical evidence or case-based analysis. Studies that were primarily theoretical, lacked practical application, or did not directly address the transformative impact of AI on urban infrastructure were excluded. Two studies were unavailable for retrieval, leaving 30 studies that fulfilled all inclusion criteria and were included in the final synthesis. The completed PRISMA flow diagram (Figure 2) visually illustrates the study selection process, showing the number of studies identified, assessed, and ultimately included at each stage.

To systematically evaluate the papers included in the review, a structured data extraction process was employed. A standardized data extraction template was developed, ensuring consistency and rigor across all selected studies. This template captured critical information, including the study objectives, the specific AI technologies or algorithms applied, the methodological approaches used, and the key results linked with the impact of AI on urban infrastructure. Also, the template included sections for identifying barriers to AI implementation, and other ethical considerations discussed. Each study's contents were

meticulously reviewed and extracted, highlighting how AI technologies are being utilized in various domains.

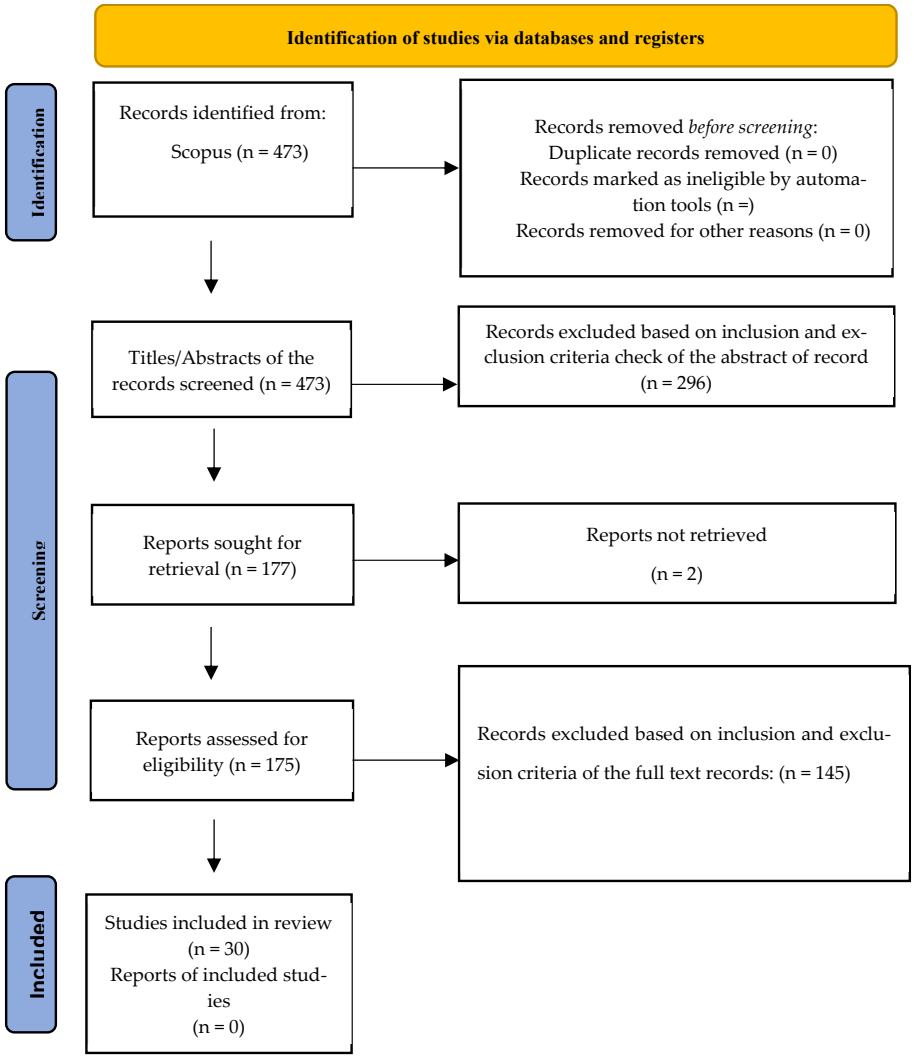


Figure 2. PRISMA framework for analysis.

After data extraction, a narrative synthesis approach was employed to organize and interpret the findings. Studies were thematically grouped according to the three focal domains, which are predictive maintenance and resource optimization, traffic and mobility management, and public participation with ethical considerations. Through qualitative analysis, common patterns and trends were identified, offering critical insights into AI's transformative potential in reshaping urban infrastructure management.

In addition to evaluating technological impacts, the synthesis also included an analysis of the ethical and social challenges of AI deployment in urban settings. This involved reviewing how AI technologies affect issues like data privacy, algorithmic transparency, and equitable access to infrastructure services. By considering these broader implications, the review provides an all-inclusive assessment of how AI can be efficiently integrated into urban systems while addressing potential risks and disparities. The narrative synthesis process helped to collate a wide range of evidence into cohesive themes, providing a detailed insight about the present state of AI-driven infrastructure management and underscoring fields for future research and development.

4. Results

4.1. Summary of Findings

The descriptive analysis covered 30 selected studies, classified based on geographical distribution, publication year, study type, research approach, and AI technologies utilized. This categorization offers a comprehensive overview of the existing research landscape on AI-driven smart infrastructure management within the context of urban planning. The geographical analysis of the reviewed studies reveals a notable regional imbalance. Most research originates from Asia, with India contributing the highest number of studies. This is followed by moderate representation from Europe and North America. The Middle East and Oceania also show some presence, while contributions from Africa and South America remain limited. This uneven spatial distribution highlights the dominance of technologically advanced regions in AI-driven urban planning research and underscores the need for more inclusive, cross-regional collaboration to ensure equitable knowledge production and application in sustainable urban development. (Figure 3).

Similarly, the temporal distribution of the reviewed studies (Figure 4) indicates a clear upward trend in research activity on AI-driven urban infrastructure management. While only isolated studies appeared in 2018 and 2022, publications have grown rapidly since 2023, peaking in 2025 with fifteen studies. This sharp increase reflects the growing academic and practical interest in applying artificial intelligence to sustainable smart infrastructure management for urban planning, particularly following global policy initiatives emphasizing digital transformation and sustainability.

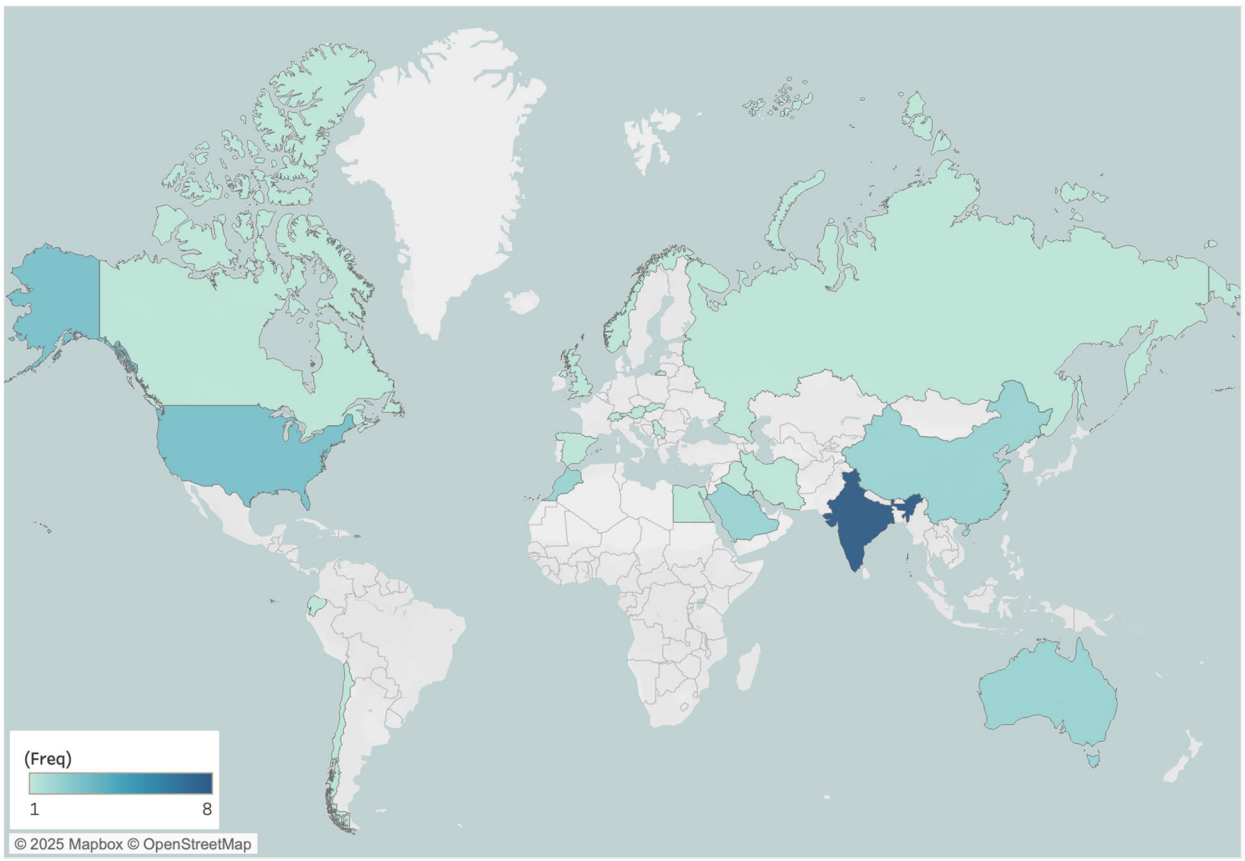


Figure 3. Geographical distribution of reviewed studies.

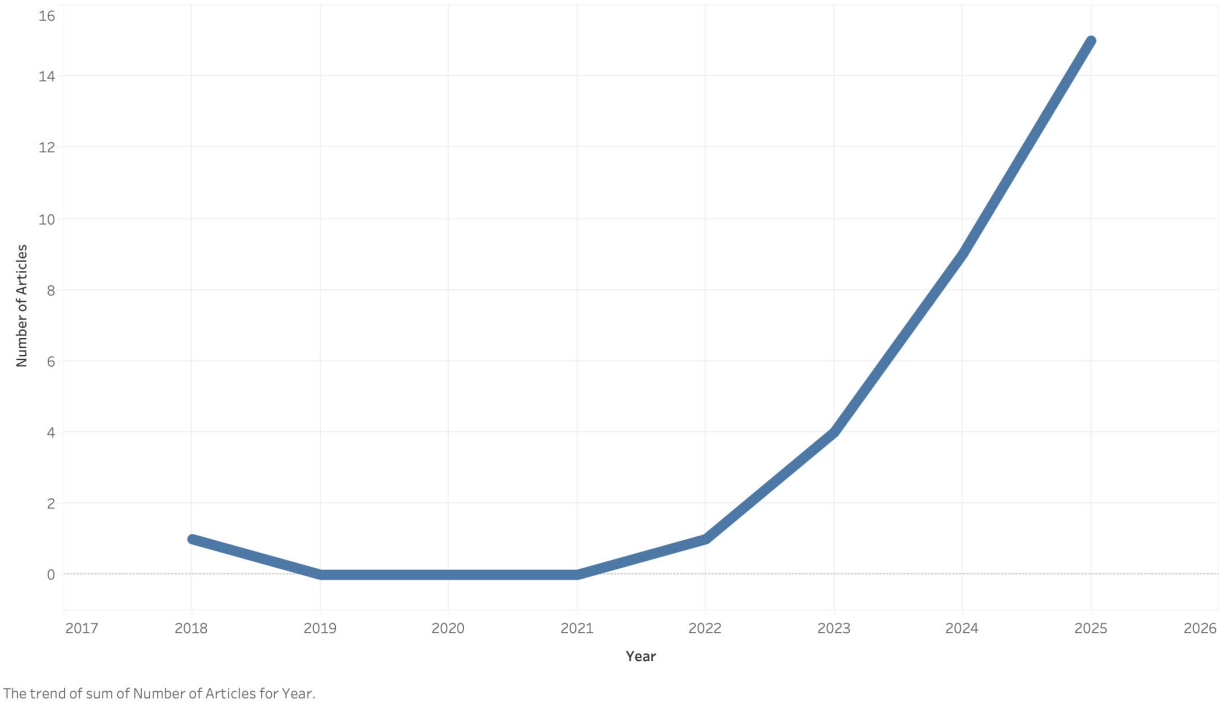


Figure 4. Number of studies published per year.

The reviewed literature comprises studies categorized into three main document types: 12 conference papers, 11 journal articles, and 7 review papers (Figure 5). Conference papers represent the largest share, reflecting the rapidly evolving and experimental nature of AI applications in urban infrastructure research. The analysis of research approaches reveals that quantitative methods dominate the reviewed studies (10 articles), followed by qualitative (7) and conceptual frameworks (6). Mixed-methods designs (4) and case studies (3) are less common but provide valuable contextual insights. This distribution suggests a strong preference for data-driven and computational approaches in AI-based urban infrastructure research, reflecting the field's emphasis on measurable performance outcomes (Figure 6).

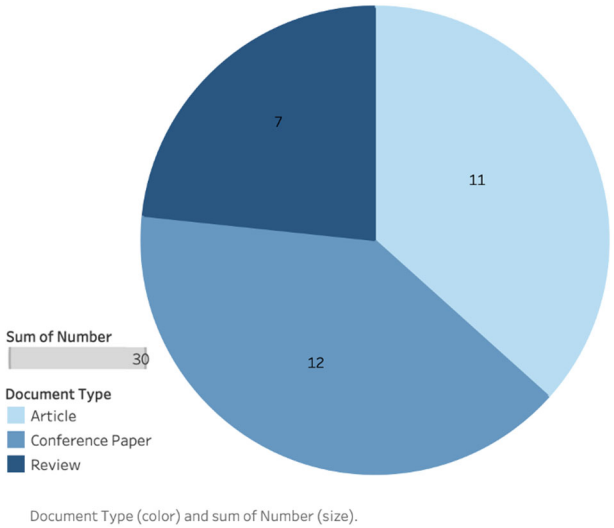


Figure 5. Document type distribution of the reviewed studies.

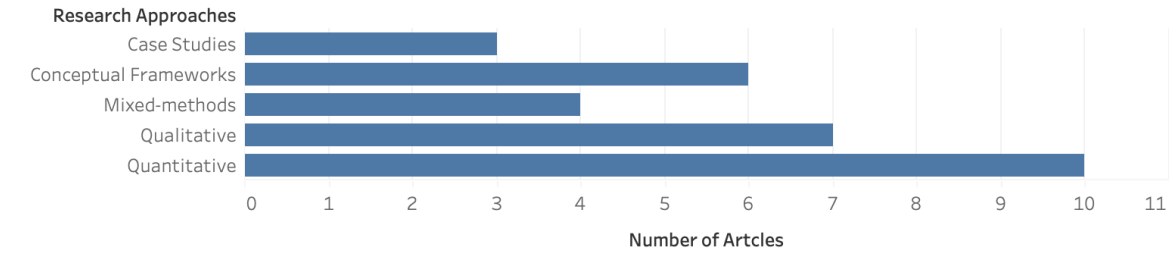


Figure 6. Research approaches adopted across the reviewed studies.

The analysis of AI technologies shows that machine learning dominates the reviewed studies (20 articles), followed by computer vision (10) and the Internet of Things (9). Deep learning (6), sensors (5), and natural language processing (3) are also frequently applied. Other technologies, such as digital twins, reinforcement learning, and graph neural networks, appear in a smaller number of studies (Figure 7). This distribution indicates that the field remains primarily driven by data-centric and sensor-based approaches, while emerging technologies such as large language models, blockchain, and VR/AR are beginning to gain attention in the context of urban infrastructure management.

To conclude the descriptive analysis, the word cloud visualization (Figure 8) offers an overview of the most recurrent concepts and themes identified across the thirty reviewed studies, offering a concise visual synthesis of AI-driven urban planning research. The most prominent terms such as artificial intelligence, smart city, urban planning, and sustainability reflect the crucial focus of the literature on integrating AI into urban infrastructure. These terms align directly with the three analytical themes identified in this review: (1) predictive maintenance and energy optimization, (2) traffic and mobility systems, and (3) public participation with ethical considerations. The visualization highlights the multidimensional nature of AI applications in urban contexts, encompassing technical efficiency, sustainable infrastructure, and inclusive governance.

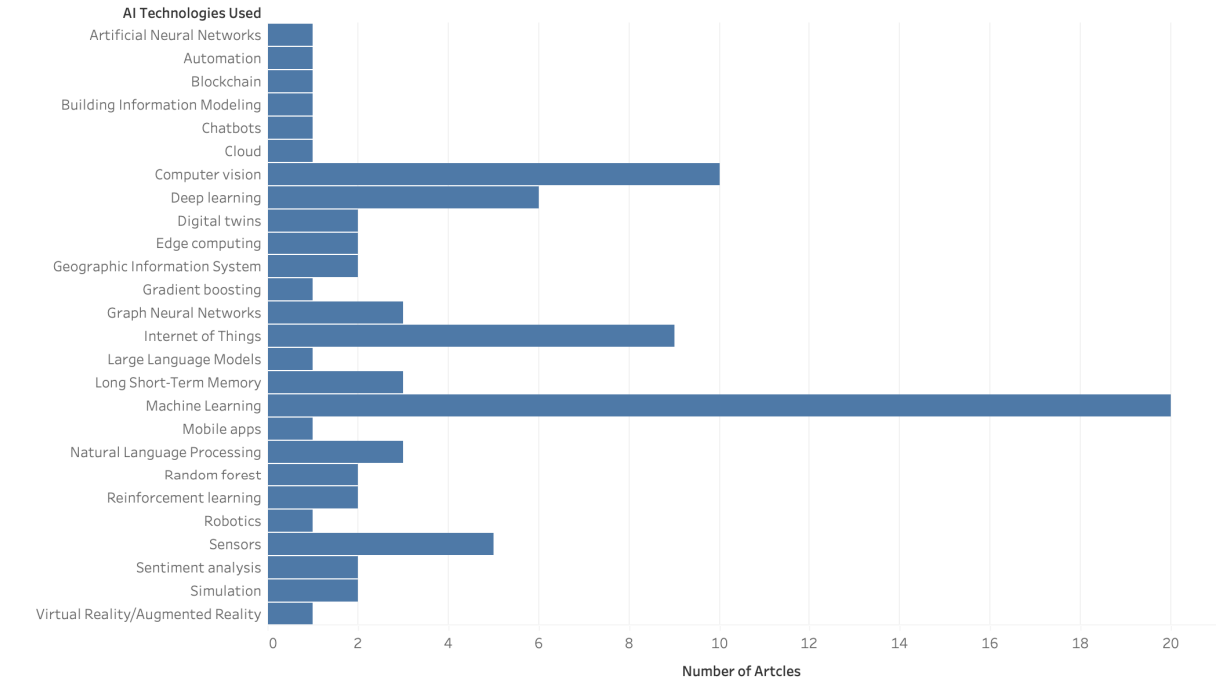


Figure 7. AI technologies across the reviewed studies.



Figure 8. Key topics addressed in the paper.

Accordingly, the thirty selected studies on smart infrastructure management for urban planning were categorized into three thematic areas, with ten studies contributing to each. The detailed findings of these thematic clusters are examined in the following sections. It is worth noticing that while the research papers were grouped under these three main categories, several studies overlap across multiple themes, reflecting the interdisciplinary nature of AI applications in urban planning.

4.2. Predictive Maintenance and Optimization of Resources and Energy

This section synthesizes findings from ten studies that examined how (AI) predictive maintenance and energy optimization, as identified across included in this review. These studies explore and machine learning (ML) improve the efficiency, reliability, and sustainability of urban infrastructure systems. The results emphasize the growing role of AI in optimizing maintenance operations, reducing energy consumption, and supporting data-driven decision-making for smart and resilient cities. Advanced computer vision techniques, particularly YOLOv9, show promise for automated road defect detection and maintenance planning [40]. Machine learning approaches achieve impressive performance metrics, with Random Forest, LSTM, and Gradient Boosting algorithms reaching up to 92% accuracy in predictive maintenance applications, while reducing costs by over 30% with an increase of 25% in uptime of operations [41].

Energy optimization represents a critical application domain, with sophisticated forecasting models achieving high precision in urban energy management [39]. Street lighting infrastructure benefits from advanced temporal modeling, where Temporal Fusion Transformers (TFT) outperform other approaches in predictive accuracy [55]. Machine learning models play a central role in achieving these optimizations. These algorithms have shown strong capabilities in capturing complex patterns in energy consumption data, supporting accurate forecasting and efficient energy distribution [39]. The TFT further enhances these outcomes by integrating contextual and temporal information to predict system faults with higher precision [55]. Similarly, predictive models for energy devices achieve notable accuracy levels, reaching about 87% in identifying potential failures up to two weeks in advance [56].

Smart grid integration emerges as a transformative application, with AI and 5G technologies enabling comprehensive energy system optimization [57]. Similarly, hydrogen energy systems particularly benefit from deep learning approaches for grid management [55]. Environmental sustainability gains prominence through AI-driven systems achieving 15–20% energy savings in cities like Amsterdam and Singapore [58]. Broader applications demonstrate substantial impact, with reported 40% energy reductions in data centers and 20% waste cost reductions [51].

The economic impact of these integrated systems extends beyond immediate cost savings. Cities implementing AI-driven predictive maintenance report a 35% increase in maintenance efficiency through reduced downtime and optimized resource allocation [41]. Furthermore, these systems contribute to broader sustainability goals by reducing carbon emissions through improved energy utilization [55] and supporting long-term energy sustainability [57]. Collectively, the evidence indicates that predictive maintenance and resource optimization operate as interconnected components within comprehensive smart city ecosystems, driving both economic efficiency and environmental resilience.

Citizen engagement through macro-scale digital twins has emerged as an innovative approach to infrastructure governance, collecting stakeholder feedback and utilizing segmentation and classification algorithms to support participatory management and infrastructure maintenance [35]. However, significant challenges persist, including ethical considerations, data privacy concerns, and IoT integration complexities in urban planning contexts. Despite these challenges, the collective evidence suggests that AI-driven infrastructure management can fundamentally shift urban systems from reactive to proactive approaches, enabling more efficient resource utilization and sustainable urban development.

Generally, this review demonstrates that AI-driven predictive maintenance and energy optimization frameworks are significantly reforming the monitoring, management, and sustainability of urban infrastructure systems. By combining advanced forecasting, computer vision, and deep learning models, these applications have proven their capacity to reduce operational costs, extend asset lifecycles, and enhance environmental efficiency. Table 3 below presents themes represented in the quoted studies regarding predictive maintenance.

Table 3. Predictive maintenance and optimization of resources and energy.

Authors and Year	Aim	Main Findings	Limitations	Ethical or Social Implications	Policy Implications
Chaymae et al. [57]	To examine how AI and 5G are integrated into smart city energy systems and identify key challenges and opportunities.	AI and 5G integration support real-time monitoring and automation in energy systems but faces issues of data heterogeneity, interoperability, and uneven infrastructure access.	Lack of standardization, cybersecurity risks, and limited attention to sustainability, especially the energy use of AI and the carbon footprint of 5G.	Challenges include algorithm transparency, data privacy, equitable access to infrastructure, and environmental impacts related to energy consumption and carbon emissions.	Promote scalable, energy-efficient AI–5G solutions, enhance data-sharing standards, and embed sustainability in smart city planning.
Majnoon and Saifoddin [39]	To develop an AI-based model for accurate energy forecasting in urban environments.	AI models effectively predict electricity consumption, showing strong potential for improving energy efficiency.	Dependence on historical data limits adaptability to changing socio-economic conditions.	The study advances sustainable policy insights by promoting data-driven energy decisions but raises equity and accessibility concerns in energy planning.	Encourage adoption of AI-based forecasting models in city energy planning and integrate socio-economic indicators to improve policy relevance and sustainability outcomes.
Whig et al. [56]	To investigate the transformative impact of Artificial Intelligence (AI) on smart city infrastructure, particularly in energy management and optimization.	AI-driven solutions significantly enhance energy efficiency and reduce consumption while improving traffic management, public safety, and waste management. The integration of AI contributes to greater urban infrastructure efficiency and sustainability.	Ethical and governance challenges persist, including the lack of clear regulatory frameworks, data privacy concerns, and limited transparency in AI system deployment.	The study highlights ethical issues related to responsible AI use in smart cities particularly fairness, accountability, and inclusivity in managing energy and infrastructure data.	Development of governance frameworks that ensure transparency, data protection, and equitable access, while promoting collaboration among public and private sectors for sustainable and inclusive AI integration.

Table 3. *Cont.*

Authors and Year	Aim	Main Findings	Limitations	Ethical or Social Implications	Policy Implications
Khedr and Abdelaziz [40]	To examine the use of robotic and AI systems for predictive maintenance in smart cities by detecting infrastructure faults.	The robotic system accurately detected cracks and potholes using YOLOv9 and Roboflow 3.0 Object Detection (Fast), though performance was limited by GPS errors and a small dataset.	Performance was affected by insufficient data diversity, GPS signal errors, and the lack of sensor integration for complex urban conditions.	Autonomous robots in public areas raise privacy and data security concerns, requiring transparent data management.	Improve dataset diversity, integrate additional sensors for accuracy, and establish clear data governance standards.
Panwar et al. [55]	To enhance energy production, distribution, and consumption through AI technologies such as deep learning and machine learning.	AI integration improves energy efficiency and sustainability by enabling predictive maintenance, demand-side management, and grid optimization. Deep learning models enhance resource allocation and grid resilience but face data security and interoperability challenges.	Key limitations include data privacy and security risks in AI-powered grid systems, computational complexity, and restricted data accessibility.	Ethical concerns involve data privacy, cybersecurity, and potential workforce shifts in the energy sector. Socially, the study highlights the need for collaboration between regulators and industry to ensure equitable adaptation.	Promote grid modernization and data protection regulations. Encourage investment in hydrogen infrastructure and develop frameworks for secure AI integration in energy systems.
Abdeen et al. [35]	To explore technologies for developing a Citizen-Centric Digital Twin (CCDT) to enhance urban infrastructure governance and citizen participation.	The study identified open-source platforms, remote sensors, and crowdsourcing as the most common data acquisition methods for CCDT. Key technologies such as WebGL and Cesium were used to support 3D visualization, data analysis, and citizen engagement in governance systems.	Challenges include data reliability, interoperability, and limited access to IoT sensors. Concerns also exist regarding data quality and consistency in volunteered geographic information (VGI) technologies.	The study highlights ethical implications related to privacy, democracy, and citizen participation through digital platforms, emphasizing the need for responsible data use in governance systems.	Recommend implementing digital platforms and data-sharing technologies to strengthen democratic participation, transparency, and collaboration in infrastructure governance.

Table 3. *Cont.*

Authors and Year	Aim	Main Findings	Limitations	Ethical or Social Implications	Policy Implications
Zehouani et al. [58]	To explore the concept of "Environmental AI" as a framework for sustainable infrastructure transformation.	Promotes shifting from technocentric approaches to ecologically and socially responsible AI. Stresses the need for ethical and legal frameworks supporting sustainable deployment.	Notes privacy and security risks, high energy use, electronic waste, and limited system adaptability.	Highlights concern about social equity, inclusivity, and the environmental impact of large-scale AI models.	Calls for integrating ethical and environmental standards into AI policies and fostering cross-sector collaboration for sustainable and fair development.
Sekar and Raja [41]	To assess the role of machine learning-based predictive maintenance in enhancing smart city infrastructure performance.	Predictive maintenance achieved up to 92% accuracy, reducing maintenance costs by about 30% and increasing operational uptime by 25%.	Raises concerns about data privacy, high infrastructure costs, and the need for specialized technical expertise.	Involves issues of privacy, cost distribution, and workforce readiness for AI-driven maintenance operations.	Encourages stronger coordination between public and private sectors to address privacy challenges, infrastructure costs, and skill development for sustainable technology adoption.
Cina et al. [51] (2025)	To explore how AI-powered predictive models support sustainable urban development through infrastructure optimization, energy management, and environmental monitoring.	AI models enhance sustainability by improving infrastructure and energy efficiency but rely on high-quality data and effective long-term implementation.	Challenges include data incompleteness, temporal inconsistencies, and algorithmic bias that may lead to unintended environmental and social outcomes if not addressed.	Raises concerns over privacy, fairness, and transparency in AI-driven urban systems, emphasizing the need for accountable and inclusive frameworks.	Promote data infrastructure, ethical collaboration, and holistic implementation strategies ensuring equity and citizen participation.
Sudharson et al. [59]	To explore predictive maintenance solutions for urban street lighting using advanced machine learning models such as TFT, TADA, and GNNs.	The TFT showed the highest accuracy in anomaly detection for time-series data, but TADA and GNNs were less effective in managing complex fault scenarios.	Current smart lighting systems still depend on reactive maintenance methods and are challenged by model precision in dynamic environments.	Enhance urban safety and supports sustainability goals through accidents reduction and the promotion of secure and efficient lighting systems.	Encourages integrating AI and IoT into urban lighting to align with UN SDGs 9 and 11, recommending predictive systems that combine IoT sensors and machine learning to foster reliability and energy efficiency.

4.3. Traffic and Mobility Management

This section presents the main findings across the ten including studies, AI and machine learning (ML) applications demonstrate substantial progress in advancing intelligent traffic management and sustainable urban mobility. AI-driven prediction and detection models show consistently high performance in traffic forecasting and incident identification. Deep learning and hybrid frameworks achieved average prediction accuracies of 91.3% and 92% [44,52], while edge-deployed detection systems reported mean average precision of 87% and F1 scores of 80% for accident identification [60]. Likewise, YOLO-based vehicle detection and counting systems have attained 95% precision and notable efficiency gains using inexpensive edge computing methods [43]. These studies underline AI's role in facilitating granular, adaptive, and data-driven traffic management processes.

Complementary research emphasizes AI's capacity for real-time optimization of urban traffic systems. Graph Neural Network and Simulated Annealing approaches reduced congestion by 25% and travel times by 18% [42], while solar-powered, IoT-integrated systems achieved a 17.2% reduction in waiting times, 22.5% improvement in emergency response, and a 15.8% decrease in carbon monoxide emissions at costs below USD 3000 [61]. Continuous routing optimization frameworks and adaptive ML models further enhance operational efficiency [44,62], reflecting a shift from reactive to proactive urban management paradigms.

Sustainability and social integration also emerge as central themes. Studies on smart infrastructure report environmental benefits and improved public engagement through multimodal systems, digital twins, and renewable energy integration [34,62]. Participatory and ethical approaches achieved up to 77.9% citizen approval and a 30% reduction in traffic offenses [63], which underscores the role of transparency and inclusiveness.

However, eight studies consistently identify barriers to large-scale implementation, including data quality, interoperability, computational demands, stakeholder resistance, and governance complexity [60,61]. Resource limitations, privacy concerns, and contextual variability further constrain generalizability. While AI-based traffic and mobility systems show transformative potential, widespread adoption requires integration of technological innovation with institutional readiness and societal trust. Table 4 below presents themes represented in the quoted studies regarding traffic and mobility management.

Table 4. Traffic and mobility management.

Authors and Year	Aim	Main Findings	Limitations	Ethical or Social Implications	Policy Implications
Saleh [64]	To propose an ethical framework for AI use in autonomous vehicles to guide policymakers and industry stakeholders.	Identifies ethical challenges in AI-driven transportation, including employment impact, algorithmic bias, and moral decision-making; proposes a structured ethical framework for AI integration.	Algorithmic bias, lack of adaptability to unforeseen scenarios, and difficulty embedding ethics into AI systems.	Highlights social concerns such as job displacement, privacy risks, and fairness; stresses transparency and responsible data use.	Calls for international collaboration to establish unified ethical standards and safety regulations, ensuring transparent and inclusive AI governance.
Mohana et al. [43]	To examine and evaluate AI tools for vehicle detection and counting.	To evaluate AI models for detecting and counting vehicles in smart traffic systems.	Current methods perform poorly under complex weather conditions and rely mainly on fixed cameras.	Addressed privacy concerns and emphasized social benefits such as reduced congestion, improved road safety, and enhanced driver awareness.	Encourages the development of intelligent vehicle systems and the use of traffic data to support infrastructure planning and congestion reduction.
Vasilieva et al. [63]	To develop a framework for engaging urban communities in the design of intelligent transport systems and improving infrastructure regulation.	Participatory design improved road safety and awareness. Real-time information tools were most approved, while automated violation systems were least accepted. Offenses fell by 30% despite low public acceptance.	Public perception of intelligent transport systems as restrictive, differing views between pedestrians and drivers, and limited approval for automated monitoring tools.	Highlighted the importance of participatory design in increasing citizen trust, acceptance of smart technologies, and satisfaction with accessibility and environmental aspects.	Promote citizen participation and transparent information sharing to enhance trust and road safety.
Adewopo & Elsayed [60]	To develop a real-time vision-based accident detection system using edge IoT devices to enhance traffic safety in smart cities.	The AI model achieved 87% accuracy in accident detection by integrating visual and motion data, offering an efficient lightweight solution for smart city surveillance.	Difficulties distinguishing slow-moving traffic from accidents, sensitivity to environmental factors, and limited generalization due to training data diversity.	Addressed privacy and fairness issues in real-time surveillance and highlighted the social importance of accessible accident-prevention systems.	Promote intelligent transport systems using connected-vehicle technology; implement vision-based detection on edge IoT devices and improve lighting and monitoring to reduce traffic fatalities.

Table 4. *Cont.*

Authors and Year	Aim	Main Findings	Limitations	Ethical or Social Implications	Policy Implications
Zhu et al. [43]	To address urban planning challenges using a hybrid AI approach combining Graph Neural Networks (GNNs) and Simulated Annealing (SA).	The hybrid GNN-SA model improved planning efficiency, reducing congestion by 25%, travel time by 18%, and infrastructure costs by up to 22%.	High computational complexity and reliance on high-quality data affect performance and scalability.	Highlighted data privacy and fairness concerns in AI-driven planning, emphasizing ethical design and collaboration.	Invest in real-time data analytics and integrate AI optimization for adaptable, inclusive urban policy frameworks.
Ansa et al. [61]	To develop an AI-driven smart traffic management system powered by renewable energy to reduce congestion and pollution in urban areas.	Introduced a renewable energy-based AI system that optimized traffic flow, prioritized emergency vehicles, and reduced emissions. Pilot tests reported a 17–22% reduction in fuel use and emissions, promoting energy savings and sustainability in smart cities.	High infrastructure and setup costs, complexity in large-scale deployment, and the need for system adaptation to different urban environments.	Promotes sustainability, air quality improvement, and alignment with UN Sustainable Development Goals (SDGs), highlighting social benefits through reduced pollution and improved public health.	Integrate AI with renewable energy in transport planning, prioritize high-need areas for gradual implementation, and adopt modular, low-maintenance infrastructure for cost-effective scalability.
Ismaeel et al. [52]	To enhance traffic prediction and management in smart cities using radial basis function (RBF).	The deep RBF model achieved 91.3% accuracy with improved precision, recall, and lower error (MAE = 3.13), showing strong potential for intelligent traffic forecasting.	Limited generalization across cities and conditions; further validation is needed for scalability and adaptability.	Improved road safety, reduced congestion and emissions, and higher efficiency in resource use through accurate predictions.	Adopt deep RBF networks to enhance forecasting accuracy and management strategies tailored to specific urban contexts.

Table 4. *Cont.*

Authors and Year	Aim	Main Findings	Limitations	Ethical or Social Implications	Policy Implications
Harisha et al. [44]	To develop an AI-based traffic prediction system using machine learning for accurate forecasting.	Developed an AI model combining historical, weather, and social media data to predict traffic with 92% accuracy, reducing congestion and improving mobility through adaptive real-time modeling.	High computational and scalability demands, limited data quality, and continuous update need restrict real-time performance.	Enhanced urban mobility, reduced congestion, and supported sustainability while emphasizing fair data handling and equitable access to technology.	Apply machine learning for traffic forecasting, integrate IoT and autonomous vehicle data, and invest in edge or cloud computing for efficient large-scale data processing.
Louati [62]	To propose a sustainable traffic management framework for AlKhari, Saudi Arabia, using AI and ML to improve safety, energy efficiency, and congestion reduction.	AI models effectively forecast traffic accidents and energy consumption, enabling proactive management and improving energy-efficient routing.	Dependence on historical data limits adaptability to unpredictable conditions; data quality issues and scalability challenges in larger cities.	Supports accident reduction, cleaner environments, and enhanced public engagement for sustainable mobility; promotes environmentally responsible transport choices.	Align AI initiatives with Saudi Vision 2030 goals, integrate advanced models for sustainable urban planning, and ensure strong data governance and privacy protection.
Yusuf et al. [34]	To review the state of sustainable urban mobility, identify challenges and knowledge gaps in digital twins and AI, and propose strategies for sustainable mobility planning.	Highlighted the role of digital twins, AI, and Big Data in improving urban mobility and governance through integrated automation, electrification, and data-driven public transit frameworks.	Overemphasis on technically oriented studies may limit interdisciplinary perspectives.	Addressed inequalities in mobility access, lack of transparency in AI policies, and the need for citizen participation and privacy-preserving governance.	Promote regulatory frameworks for AI and Big Data adoption, participatory urban planning, and transparent, ethical use of digital twins in mobility systems.

4.4. Public Participation and Ethical Considerations

This analysis of ten studies reveals critical insights into how public participation and ethical frameworks shape AI implementation in urban planning, highlighting both opportunities and persistent challenges. Digital platforms consistently emerge as key enablers of community engagement. Generative AI applications demonstrate promise, automating visualization to lower participation barriers and improve engagement, especially for disengaged groups [27]. Digital agents and chatbots enhance stakeholder engagement, with a specific focus on inclusivity for older adults [65]. However, studies reveal significant variation in approaches, with collaborative models in Helsinki and Copenhagen contrasting sharply with more technocratic implementations in Dubai and Tokyo [25].

Privacy is increasingly becoming the most cited ethical concern across the reviewed literature [25,66], followed by transparency [36,67] and accountability [67,68]. Regulatory frameworks, like the AI Act by the European Union, offer governance benchmarks, while participatory design processes explicitly address equity concerns and counteract green gentrification [37]. Studies emphasize combining traditional ethical principles with AI-specific considerations like explainability [65].

Despite positive outcomes in engagement and inclusiveness, persistent barriers include the digital divide, algorithmic bias, and regulatory gaps [37]. Critical gaps emerge between theory and practice, as demonstrated by Vienna's traffic system improving efficiency while failing to address visually impaired users' needs [45]. Studies consistently call for robust frameworks ensuring transparency, ongoing oversight, and balancing digital with traditional participation methods to prevent exclusion of marginalized groups [25]. As a result, these findings suggest that AI can democratize urban governance when ethical, transparent, and participatory frameworks are institutionalized alongside technological innovation. Table 5 below represents themes of publication, participation and decision-making.

Table 6 summarizes the key benefits and challenges of AI-driven smart infrastructure management across the three thematic domains of this review: predictive maintenance and resource/energy optimization, traffic and mobility management, and public participation with ethical considerations. While each domain demonstrates substantial potential for improving efficiency, sustainability, and inclusiveness, the analysis also highlights recurring barriers such as data quality issues, integration with legacy systems, and ethical concerns.

Table 5. Public participation and ethical considerations.

Authors and Year	Aim	Main Findings	Limitations	Ethical or Social Implications	Policy Implications
Ryan and Gutman [65]	To explore how digital agents and chatbots enhance stakeholder engagement and inclusivity for older adults, and to address ethical concerns in AI-supported smart cities by combining traditional ethics with AI principles.	AI-supported smart cities improve management and services for older adults but risk excluding them due to digital divide and lack of transparency. A combined ethical framework is needed to ensure dignity and autonomy.	Current research is mainly conceptual and lacks empirical studies on AI's impact on older adults in smart cities.	Highlights risk of exclusion, privacy issues, and reduced autonomy; calls for balancing efficiency with dignity and fairness for older adults.	Design inclusive AI-supported smart city strategies that protect dignity and autonomy and avoid socio-economic inequities.
Sepehr [45]	To examine how AI integration in urban governance shapes urban spaces and political configurations.	The Pedestrian Traffic Light system overlooks the needs of diverse urban residents, particularly the visually impaired. There is a need for more inclusive AI-centric urban projects.	Current AI systems show limited inclusivity, often neglecting diverse user needs in urban governance.	Emphasizes inclusivity, equity, and the need to democratize AI-driven urban spaces through socially aware design.	Adopt inclusive and participatory AI approaches in urban planning, fostering accessibility and representation for all residents.
Cath [36]	To explore ethical, legal, and technical challenges in AI governance and propose inclusive and effective governance frameworks.	AI governance remains inadequate, with narrow legal approaches and oversimplified technical solutions; broader, inclusive frameworks are needed to address ethical, legal, and regulatory challenges.	Existing legal and technical solutions are limited in scope and fail to address broader social implications.	Focus on fairness, transparency, privacy, and the impact of AI on human rights and democracy, especially power concentration in corporations.	Promote human rights–based AI governance through inclusive stakeholder participation, accountability, and equitable regulation.
Kresović and Vukmirović [66] (2025)	To explore how AI technologies can enhance smart governance in Serbia by promoting citizen participation through e-governance and mobile applications.	AI improved governance, mobility, and sustainability, but challenges remain in infrastructure, data security, and regulation.	Emphasizes the need for robust digital infrastructure and strong data protection measures to safeguard citizen privacy and prevent cyberattacks.	Highlights the need for smart cities to address social and environmental challenges beyond technology, emphasizing data security, privacy, digital inclusion, and legal frameworks for ethical governance.	Develop AI-based smart city frameworks, invest in digital infrastructure, ensure data security, and promote inclusive citizen participation.

Table 5. *Cont.*

Authors and Year	Aim	Main Findings	Limitations	Ethical or Social Implications	Policy Implications
Villegas-Ch et al. [68] (2025)	To design and validate a decentralized computational architecture for urban governance.	The decentralized system achieved 95–81% accuracy and over 83% blockchain validation with stable performance.	Limited scalability; testnet may not reflect real-world network conditions.	Enhance transparency and accountability through blockchain validation.	Promote decentralized control and ensure blockchain-based traceability.
Zhao [37] (2025)	To analyze how Shenzhen uses smart technologies for green urban planning and identify fair, replicable strategies.	Inclusive participatory frameworks integrate community input and enhance equitable access to smart infrastructure.	Data bias, algorithmic limitations, and equity gaps.	Highlights risk of inequality and "green gentrification" caused by technological advantages.	Promote equitable financing, anonymized IoT data frameworks, and standardized PPP models for fair smart city deployment.
Lartey and Law [67]	Critically assess AI adoption in urban planning/governance; examine socio-political, ethical, and governance issues; propose equitable, inclusive frameworks.	Need for inclusive, participatory frameworks balancing efficiency and equity.	Mainly theoretical, lacks empirical evidence.	Gaps persist in transparency, accountability, and inclusivity; urban AI often overlooks community participation and equity, risking unequal outcomes.	Adopt inclusive AI governance frameworks with continuous stakeholder engagement.
Lützelschwab et al. [27]	Evaluate CitizenVision, a GenAI app for visualizing and sharing urban redesign ideas, assessing user engagement.	CitizenVision improved user experience and participation; GenAI tools enhanced clarity and visualization.	Lacks iterative features; needs integration with other tools and large-scale testing.	Promotes inclusivity and accessibility in planning by reducing participation barriers.	Integrate GenAI tools with complementary features; support collaborative design and large-scale testing.
Sanchez et al. [25]	Identify key ethical concerns in AI-driven urban planning, focusing on bias, transparency, accountability, privacy, and misinformation.	AI can enhance participation but raises ethical issues related to bias, privacy, and misinformation. Emphasizes inclusive, transparent frameworks.	Limited literature on AI ethics in urban planning; evolving technologies require ongoing research and adaptive strategies.	Bias and privacy risks can reinforce inequality and marginalization; planners must uphold fairness and justice.	Promote public engagement, ethical standards, transparency, inclusivity, and data privacy

Table 5. *Cont.*

Authors and Year	Aim	Main Findings	Limitations	Ethical or Social Implications	Policy Implications
Berigüete et al. [26]	Explore how emerging technologies enhance citizen participation in sustainable urban planning and promote inclusion.	Digital tools like 5G and online platforms increase citizen engagement and align with sustainability goals.	Limited case coverage; lacks analysis of socio-economic and regional disparities, especially in the Global South.	Concerns about privacy, equity, and access; highlights need for inclusive participation and data accountability.	Promote transparency, inclusive decision-making, adaptive planning, and responsible data governance.

Table 6. Summary of Key Benefits and Challenges of AI-Driven Smart Infrastructure Management.

Domain	Key Benefits	Key Challenges
Predictive maintenance and resource/ energy optimization	Reduces equipment downtime and operational costs (up to 30%). Extends asset lifespan (roads, bridges, utilities). Improves energy/resource efficiency and reduces waste.	Integration with legacy infrastructure. High initial costs of AI systems. Data quality and availability constraints.
Traffic and mobility management	Reduces congestion and travel times. Lowers emissions and fuel consumption. Enhances real-time traffic monitoring and adaptive control.	Need for large-scale reliable data. Technical complexity in deployment throughout the city. Risk of uneven access among socio-economic groups.
Public participation and ethical considerations	Enhancing transparency and public engagement. Reinforces trust in infrastructure projects. Supports inclusive decision-making.	Data privacy concerns. Algorithmic bias and fairness. Digital divides among marginalized groups.

5. Discussion

5.1. Synthesis for the Findings

This study has examined the role of AI in advancing sustainable urban infrastructure across three key domains: predictive maintenance and resource optimization, traffic and mobility management, and public participation with ethical considerations. The evidence across the reviewed studies indicates that AI plays a pivotal role in improving the reliability, efficiency, and resilience of urban systems. In the context of predictive maintenance, AI applications can help anticipate infrastructure failures and reduce operational disruptions, achieving cost savings of up to 30% in urban transportation and other public infrastructure sectors [69]. Consistent findings across multiple studies highlight AI's contribution in enhancing service continuity and reducing maintenance burdens in high-density urban environments [38,70,71].

The synthesis of findings from the reviewed studies indicates that AI-driven smart infrastructure management is reshaping urban planning by enabling predictive maintenance and energy optimization. While most studies converge on the effectiveness of AI in enhancing efficiency and sustainability, they differ in methodological focus and application domains. Deep learning and ensemble algorithms, including LSTM networks, Random Forest, and Gradient Boosting, demonstrate superior performance in predictive maintenance [41]. Similarly, computer vision approaches such as YOLOv9 automate road defect detection with high precision [40], and Temporal Fusion Transformers outperform other temporal models in optimizing street lighting [55].

Comparatively, energy optimization studies emphasize broader system-level transformations. Forecasting frameworks achieved high accuracy in urban energy prediction [39], while practical implementations demonstrate measurable energy savings in cities such as Amsterdam and Singapore [58]. AI-driven systems further report efficiency improvements in data centers [52] and successful integration of 5G-enabled smart grids [57] and hydrogen energy systems [55]. These studies collectively show that while technical models achieve

strong predictive accuracy, their success varies depending on data quality, contextual adaptability, and infrastructure readiness.

From an economic perspective, AI-driven predictive maintenance systems show tangible returns, with cities achieving around 30% reductions in maintenance costs and 25–35% improvements in operational uptime due to automated fault detection and optimized scheduling [41]. These results suggest that predictive maintenance and energy optimization operate as interdependent systems that reinforce both economic efficiency and environmental resilience. Environmentally, AI-enabled frameworks contribute to sustainability goals through reductions in carbon emissions and improved energy utilization efficiency [55,57], positioning these technologies as key drivers of responsible urban growth.

In contrast, social and participatory frameworks, such as citizen-centric digital twins, emphasize governance transparency and citizen engagement by positioning residents at the core of infrastructure decision-making. These systems employ real-time feedback mechanisms, including volunteered geographic information and social sensing, to strengthen collaboration and accountability [35]. However, persistent challenges remain, particularly in data interoperability, the absence of comprehensive ethical frameworks, and the environmental impacts of AI systems related to energy consumption and electronic waste generation [58]. Across several studies, recurring barriers include data interoperability, IoT integration challenges, and ethical or regulatory uncertainties [57,58]. In general, these converging findings highlight both the technological maturity and the socio-technical challenges of AI-based infrastructure management, setting the foundation for the discussion that follows on how these systems can be more effectively and ethically integrated into sustainable urban development.

Building on these insights into predictive maintenance and energy optimization, the discussion now turns to the application of AI in transportation and mobility systems, where similar data-driven principles are reshaping urban efficiency and sustainability. AI-driven smart infrastructure management is transforming urban planning by revolutionizing traffic and mobility systems through data-driven optimization and predictive analytics. The findings of this review reveal that artificial intelligence (AI) has become a pivotal enabler of sustainable and intelligent urban mobility. The synthesis of the ten reviewed studies collectively in Table 5 demonstrate that AI-driven systems can substantially improve traffic prediction, congestion management, and environmental performance [42,52,72]. Deep learning and hybrid models such as LSTM, RBF, YOLOv3, and GNN-based frameworks consistently outperformed traditional analytical methods, achieving high predictive accuracy and operational efficiency [44,60]. These results align with previous findings emphasize that data-driven and adaptive systems transform urban transport from reactive control to proactive optimization [61,62].

However, the discussion extends beyond technical efficiency. Persistent implementation barriers such as data quality, interoperability, scalability, and governance complexity indicate that technological readiness alone is insufficient for systemic transformation [34,42]. Similarly to findings in broader smart city research, these constraints reflect the need for institutional capacity building and stronger stakeholder engagement to ensure real-world adoption of AI systems [63]. Similarly, ethical and societal dimensions are equally critical. Studies such as Saleh [64] and Vasilieva et al. [63] report public concerns regarding privacy, fairness, and autonomy, illustrating that social acceptance is essential to sustaining innovation. The proposed cyclic framework for ethical AI governance [64] and participatory design approaches [63] collectively point toward an emerging paradigm of "responsible AI urbanism," which balances technological efficiency with societal well-being.

While advancements in AI-driven mobility demonstrate the potential for efficiency and safety, broader ethical and participatory dimensions remain central to ensuring that

such technological progress aligns with social values and inclusive governance. This synthesis reveals both convergent ethical priorities and divergent participatory approaches across all reviewed studies, highlighting the complex relationship between technological innovation and democratic governance in AI-driven urban planning. Studies demonstrate remarkable consistency in ethical priorities, with privacy emerging as the predominant concern [25,36,37], followed by transparency [65,67], and accountability [67]. This consistency suggests universal recognition of core democratic values in AI governance, regardless of geographic or cultural context.

Geographic variations reveal fundamentally different approaches to public engagement. European studies emphasize collaborative, inclusive models with community oversight boards and participatory review processes [25,45] while Asian implementations adopt more efficiency-focused approaches with structured equity assessments [37]. North American studies bridge these approaches, combining digital innovation with traditional democratic principles [65].

All reviewed studies have acknowledged the potential of digital platforms in enhancing engagement through real-time feedback, generative AI visualization, and mobile applications [27,65]. However, studies consistently identify the digital divide as a critical barrier, with marginalized groups facing exclusion [25]. This creates a fundamental tension between technological advancement and inclusive participation. Studies reveal persistent disconnection between ethical frameworks and practical outcomes. While regulatory frameworks for responsible technology use provide governance benchmarks, implementation failures such as Vienna's traffic system neglecting visually impaired users demonstrate the challenge of translating ethical principles into equitable practice across diverse urban contexts [45]. These insights highlight that ethical and participatory frameworks are not secondary considerations but foundational to the legitimacy and long-term sustainability of AI-driven urban governance.

As a result, the three thematic areas predictive maintenance and energy optimization, AI-driven mobility, and public participation with ethical considerations illustrate that the success of AI in urban governance depends not only on technical efficiency but also on social legitimacy. Sustainable smart cities require aligning technological innovation with ethical accountability, transparency, and inclusive participation to ensure that digital transformation serves both urban efficiency and human well-being.

5.2. Technical and Ethical Challenges in AI Implementation

Despite its advantages, AI integration in urban infrastructure faces several technical and ethical challenges. One major barrier is the complexity of integrating AI with legacy infrastructure, especially in cities with systems that lack digital compatibility [25,72]. As this study observed, cities in emerging markets are particularly constrained by limited computational resources and aging infrastructure, which complicates the adoption of AI-driven solutions [40]. Scalability also presents challenges. High-density cities, even with robust digital frameworks, must process vast quantities of real-time data, posing a significant computational burden [60]. Consistent performance across diverse urban settings requires scalable AI systems, but high implementation costs and integration complexities remain obstacles [41].

Ethical concerns, such as privacy, algorithmic fairness, and the digital divide, further complicate AI's deployment in urban systems [37]. With AI systems handling sensitive public data, it is crucial to ensure transparency and maintain data security. Algorithmic bias and socio-economic disparities can also impact equitable access to AI-driven resources, risking the potential exclusion of marginalized communities. Ensuring that AI applications in urban planning are inclusive and accountable is essential for achieving fair, sustainable

urban growth [73,74]. A critical challenge lies in balancing the efficiency gains of AI with broader equity concerns.

While AI-driven optimization can significantly improve traffic flow, energy distribution, and resource allocation, it also risks deepening socio-economic divides if access remains unequal or if algorithmic bias persists. These ethical trade-offs underscore the tension between maximizing operational performance and ensuring fair, inclusive outcomes, highlighting the need for transparency, fairness, and accountability in AI system design [75]. Scholars further emphasize that urban automation often prioritizes efficiency while marginalizing vulnerable groups, reinforcing the importance of ethical safeguards [76]. Several studies call for ethical frameworks that explicitly integrate fairness, transparency, and inclusiveness into AI-enabled systems, ensuring that efficiency gains do not come at the expense of equity [77].

5.3. Implications for Sustainable Urban Development

The integration of AI-driven smart infrastructure management presents transformative implications for sustainable urban development, demonstrating how technology can address multiple sustainability challenges simultaneously. The reviewed studies reveal that AI implementation generates cascading sustainability benefits across environmental, economic, and social dimensions, directly supporting the Sustainable Development Goals (SDGs).

Environmental sustainability emerges as the most quantifiable benefit, aligning with SDG 7 and SDG 13. Energy optimization systems achieve 15–20% reductions in consumption, potentially preventing billions in climate-related damages [37,58]. Similarly, AI-enabled traffic management systems contribute to SDG 11 by reducing emissions by 21–25% and improving urban mobility [51]. The convergence of predictive maintenance and resource optimization extends infrastructure lifecycles up to 100 years, reducing construction-related environmental impacts and resource waste [40].

Economic sustainability aligns with SDG 8 and SDG 9 through cost reductions and improved resource efficiency. Maintenance cost decreases of 25–30%, coupled with energy savings equivalent to 1.6 TWh annually, illustrate how AI fosters economically viable pathways to sustainability [37,41]. These economic benefits can enable reinvestment in further sustainable infrastructure development, creating positive feedback loops for urban transformation.

Social sustainability contributes to SDG 10 and SDG 16 by emphasizing equity, inclusivity, and ethical governance. Despite challenges related to digital access and participation [45], participatory AI approaches and citizen-centric digital twins promote transparent, inclusive decision-making and help bridge digital divides [35].

As a result, these implications caution that achieving SDG 11 requires integrating technological innovation with ethical governance and inclusive design. AI-driven urban systems can thus serve as powerful instruments for advancing global sustainability agendas that are environmentally resilient, economically viable, and socially just.

6. Conclusions

This study reviewed thirty relevant literatures on the transformative potential of AI in urban infrastructure management, across the three domains of predictive maintenance and optimization of resources and energy, traffic and mobility management, and public participation with ethical considerations. AI generates interdependent benefits that strengthen overall urban functionality beyond improving efficiency within each domain. However, challenges such as legacy system integration, scalability, and ethical concerns exist, highlight the need for context-sensitive and well-governed implementations. Based

on the findings of this study, implications are drawn, and practical recommendations are offered to guide policymakers, practitioners, and researchers in shaping responsible AI adoption for sustainable urban futures.

For practitioners, the findings suggest that AI can be policy-driven and strategically deployed to reduce operational disruptions, optimize mobility, and foster community participation, but its success depends on strong governance and ethical safeguards. Policymakers should prioritize participatory platforms, interoperable systems, and transparent governance frameworks, while ensuring equitable access to AI-enabled infrastructure to reduce the digital divide. Governments should also establish guidelines to protect data privacy, ensure algorithmic fairness, and maintain public trust. In addition, collaboration between public and private sectors remains critical to overcoming cost barriers, particularly in emerging markets.

For researchers, the study highlights the need to evaluate AI not only for its technical effectiveness but also for its broader social, ethical, and equity implications. Future studies should focus on adapting AI to diverse urban environments, accounting for geographic and socio-economic differences, and exploring its role in building resilience to climate change, natural disasters, and socio-economic disruptions.

Finally, several ongoing challenges warrant further research. These include ensuring fair access to AI-enabled services across populations, achieving seamless integration with legacy systems that still underpin many urban services, and developing robust governance frameworks to regulate equitable and ethical adoption. Addressing these challenges will be critical for realizing the full transformative potential of AI in creating sustainable, inclusive, and resilient urban environments.

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Appendix A

Search Strategy.

TITLE-ABS ((“artificial intelligence” OR “AI” OR “machine learning” OR “deep learning”) AND (“smart infrastructure” OR “urban infrastructure” OR “urban planning”) AND ((“predictive maintenance” OR “resource optimization” OR “energy optimization”) OR (“traffic management” OR “mobility management”) OR (“public participation” OR “citizen engagement” OR “ethical considerations” OR “equity” OR “governance”))).

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